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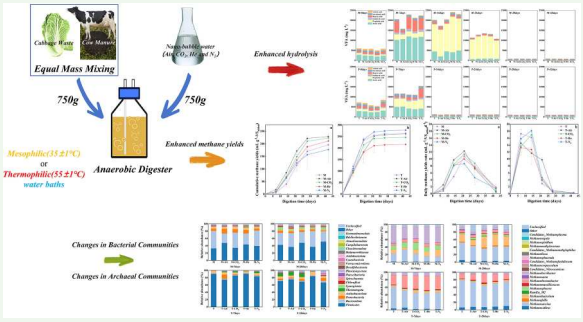
Effects of nano-bubble water on anaerobic co-digestion of cabbage waste and cow manure under mesophilic and thermophilic conditions

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ABSTRACT

The impact of four nano-bubble water (NBW) additions on the hydrolysis rate, methane yield, and microbial community of co-digestion of cabbage waste (CW) and cow manure (CM) under mesophilic and thermophilic conditions were investigated. Adding NBW under mesophilic conditions promoted hydrolysis, and the highest soluble chemical oxygen demand of the mesophilic digesters with the addition of CO_2-NBW increased by 15.86%. Methane yield in the mesophilic digesters with Air-NBW and CO_2-NBW increased by 17.54% and 14.72%, respectively. Moreover, the addition of NBW further accelerated the methane yield rate under mesophilic conditions. Due to the influence of thermophilic temperature, the impact of NBW addition on hydrolysis, methane yield, and methane yield rate in the thermophilic digesters did not differ significantly from the control. The addition of Air-NBW and N_2-NBW in the thermophilic digesters resulted in only marginal increases in methane yield, by 1.09% and 5.61%, respectively. NBW addition enhanced both the abundance and diversity of microbial communities in both mesophilic and thermophilic digesters. The addition of NBW represents a promising technological advancement for enhancing the efficiency of anaerobic co-digestion of CW and CM.



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KEYWORDS
Nano-bubble water; hydrolysis; archaeal communities; bacterial communities; methane yield

1. Introduction

As a predominantly agricultural nation, China generates an estimated 360 million tons of vegetable waste annually [1]. Cabbage waste (CW), a common food waste in China, is characterized by its high biodegradability [2]. Improper management of CW may result in significant health and environmental risks [3]. Anaerobic digestion (AD) offers an effective solution for treating CW by decomposing the organic matter in vegetables to produce biogas, while the resulting digestate serves as a valuable fertilizer [4]. The high volatile solids (VS) content (approximately 75%) and low total solids (TS) content (below 10%) in CW can lead to the accumulation of volatile fatty acids (VFA), which in turn may inhibit methanogenic activity and potentially result in the failure of the AD [3].

Co-digestion of CW and cow manure (CM) presents a promising strategy for stabilizing the AD process and enhancing methane yield [5]. Integrating CW and CM can effectively balance the water content, biodegradable organic matter, and buffering capacity within the AD [3]. Vegetable peels were combined with CM under mesophilic anaerobic co-digestion conditions, achieving a methane yield of 170 mL g^{-1}-VS_{added} [6]. However,

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when the proportion of CW in the co-digestion was too high, the alkalinity of CM was not sufficient to buffer the hydrolysis to produce VFA, leading to the VFA accumulation [3]. In response to VFA accumulation, the application of additives, pH adjustments, or biofortification techniques can effectively mitigate acid inhibition [7]. Application of additives is an effective way to optimize the stability of CW and CM co-digestion systems and improve methane yield.

Adding biochar, zeolite, powdered activated carbon, and NaOH improved the stability of AD and increased methane yield [8]. Nano-bubble technology not only enhances the stability of AD and methane yield but also facilitates better utilization of the digestate, owing to the absence of chemical additives [9]. Nano-bubble water (NBW) is characterized by an abundance of nano-sized bubbles, which exhibit superior mobility and gas solubility than distilled water (DW) [10]. During their collapse, nano-bubbles generate substantial amounts of reactive oxygen species, which in turn facilitate the hydrolysis of organic compounds [11,12]. NBW addition increased the activity of bacteria and methanogens, thereby promoting methane yield [13]. Air-NBW and O_2-NBW enhanced the activity of facultative bacteria by creating a microaerobic environment, which in turn increased the efficiency of the electron transport system, thereby reducing the accumulation of VFA [14,15]. At the same time, NBW was added to increase alkalinity, improving buffering capacity and enhancing the stability of AD [9].

The addition of CO_2-NBW into swine manure AD at 8% TS effectively reduced the accumulation of VFAs, resulting in a cumulative methane yield of 261 mL g^{-1}-VS_{added} [13]. The addition of N_2-NBW and O_2-NBW enriched the microbial communities and enhanced enzyme activity in the AD of corn straw, leading to increases in methane yield by 18.1% and 46.8%, respectively [15]. The addition of N_2-NBW to the AD of sludge increased methane yield by 29.0% compared with the control, the reason for which was mainly attributed to the enhancement of the enzyme activity of extracellular hydrolases by the addition of N_2-NBW [16]. CW is easily hydrolyzed and acidified, CM is difficult to hydrolysis but can improve buffering capacity [3,6]. Achieving enhanced methane yield from AD necessitates a close linkage between the hydrolysis and methanogenesis stages, rendering the use of NBW to augment these stages a promising approach. The effect of NBW addition on the hydrolysis and methanogenic stability of CW and CM co-digestion is not known. Meanwhile, the effect of NBW addition on the changes of microbial communities during the co-digestion of CW and CM is also unclear.

In this study, the effect of adding four types NBW (Air, CO_2, He, and N_2) to the co-digestion system on the performance of anaerobic co-digestion of CW and CM, and microbial community under mesophilic or thermophilic conditions was investigated. Enhancing a deeper understanding of NBW's role in the co-digestion of CW and CM is crucial for optimizing its application in co-digestion systems.

2. Materials and methods

2.1 Batch anaerobic fermentation experiments

2.1.1 Raw materials and inoculum

The CW was sourced from a wholesale vegetable market located near the laboratory in Lanzhou, China. The CM was sourced from a dairy farm on the outskirts of Lanzhou, where predominantly Chinese Holstein cows are bred. Prior to the experiment, the CW was crushed using a grinder and homogenized to prepare it as fermentation material, after which it was stored together with the CM in a refrigerator at 4 ± 1 °C for preservation. The inoculum used in the mesophilic and thermophilic digesters consisted of mesophilic and thermophilic seed sludge, which had been pre-adapted in the laboratory. The major characteristics of the various substrates utilized in this study are summarized in Table 1.

2.1.2 Production of NBW

The production of NBW with different gases was carried out using a nano-bubble water generator (JXWNP-AL01, Jinxiang, China). The NBW generated by this instrument featured an average nano-bubble particle size of 135.8 nm and a concentration of 537 million nano-bubbles mL^{-1} (Figure S1). Gas (Air, CO_2, He and N_2) was vented into distilled water through the generator for 20 minutes at 20 ± 2°C, followed by a 10 min stand period to obtain the NBW. During the NBW preparation process, the inlet pressure was maintained between

Table 1. Characteristics of cabbage waste, cow manure and seed sludge.

	Cabbage waste	Cow manure	Mesophilic Seed sludge	Thermophilic Seed sludge
TS (%)	5.05 ± 0.76	37.46 ± 0.71	8.62 ± 0.01	5.74 ± 0.03
VS (%)	3.98 ± 0.08	19.58 ± 0.03	5.76 ± 0.02	3.42 ± 0.01
VS/TS (%)	78.81 ± 0.48	52.26 ± 0.10	66.82 ± 0.01	59.58 ± 0.03
C (%)	43.76 ± 0.18	32.80 ± 0.24	31.25 ± 0.26	30.22 ± 0.70
N (%)	3.84 ± 0.05	2.03 ± 0.01	2.03 ± 0.02	2.24 ± 0.01
H (%)	5.19 ± 0.03	4.59 ± 0.03	4.25 ± 0.04	4.15 ± 0.10
S (%)	0.54 ± 0.01	0.59 ± 0.02	0.54 ± 0.01	0.57 ± 0.16
pH	5.71 ± 0.01	8.31 ± 0.02	7.15 ± 0.02	7.05 ± 0.01

C, N, H and S content determination is based on the dry matter of the sample.
Average and standard deviation were calculated based on three parallels.

0.4 and 0.5 MPa. After production, NBW was quickly added to the digesters under different conditions.

2.1.3 Experimental setup

Based on previous research finding, a CW-to-CM ratio of 1:1 on a wet basis has been shown to higher methane yield [3]. Consequently, the CW and CM were mixed in a 1:1 ratio on a wet basis. The mixed substrate was mixed with an equal mass of either NBW or DW, resulting in a total mass of 1500 g. The five experimental groups in the experiment were DW, Air-NBW, CO_2-NBW, He-NBW, and N_2-NBW were added, respectively. These five experimental groups underwent batch AD experiments under thermophilic or mesophilic conditions. Under mesophilic conditions, five groups were designated as M, M-Air, M-CO_2, M-He, and M-N_2. Under thermophilic conditions, the five groups were designated as T, T-Air, T-CO_2, T-He, and T-N_2. Subsequently, 500 g of mesophilic or thermophilic seed sludge was added to the corresponding digesters.

The entire digestion process was conducted in a digester with a working volume of 2 L. The digester was sealed with a specialized lid designed for AD, through which two metal tubes with an internal diameter of 4 mm were inserted. Metal tubes with silicone tubing at both ends were employed for sampling via peristaltic pumps, while metal tubes with silicone tubing extending outside the bottle were used to connect to gas bags for biogas collection. The entire experiment was temperature controlled using a water bath, maintaining the mesophilic digesters at 37 ± 1 °C and the thermophilic digesters at 55 ± 1 °C. Each morning, each digester was manually shaken for 1 min. All digesters were run in triplicate.

Following the scheduled shaking of the digester, a peristaltic pump was used to withdraw a portion of the digestate mixture. The mixture was then centrifuged at 2000g for 5 min, after which the supernatant was filtered through a 0.45-μm polyether sulphone microfibre filter. The resulting filtrate was utilized for subsequent experimental analyses. A 5 g sample of the mixed digestate was taken at a predetermined time as the microbiological sample, and stored at −80 °C until sequencing.

2.2 Analytical methods

TS, VS, pH, soluble chemical oxygen demand (SCOD), total ammonia nitrogen (TAN), and total alkalinity (TA) were analysed following the standard methods [17]. Free ammonia nitrogen (FAN) was calculated based on pH, TAN, and temperature [18]. VFA was quantified using a gas chromatograph (SP-7890 Plus, Ruihong, China) equipped with an organic acid analytical column (320 μm inner diameter, 0.50 μm film, and 30 m total length) and a flame ionization detector. The biogas volume was measured using an acid gas flow metre (LML-2, Alpha, China), while the methane concentration was assessed using a biogas composition analyser (GA5000, Geotech, United Kingdom).

Deoxyribonucleic acid (DNA) was extracted from the digestate at specific intervals, followed by microbial abundance and diversity analysis using high-throughput sequencing to assess differences in microbial communities across co-digesters treated with different types of NBW under different temperature conditions. The V3-V4 hypervariable regions of the bacterial 16S rRNA gene were amplified using the primers 341F (5′-CCTACGGGNGGCWGCAG-3′) and 806R (5′-GGACTACHVGGGTATCTAAT-3′). The V4-V5 hypervariable regions of the archaeal 16S rDNA gene were amplified using the primers Arch519F (5′-CAGCMGCCGCGGTAA-3′) and Arch915R (5′-GTGCTCCCCCGCCAATTCCT-3′). In brief, DNA was extracted from each digestate sample using the HiPure Soil DNA Kit (Magen, Guangzhou, China), followed by sequencing on the Illumina MiSeq platform (AxyPrep DNA Gel Extraction Kit, Axygen Biosciences, Union City, CA, U.S.). The raw microbial sequencing data were deposited in NCBI (https://www.ncbi.nlm.nih.gov/), with accession numbers PRJNA1043753 for bacteria and PRJNA1043758 for archaea.

2.3 Statistical analysis

Data analysis was performed using SPSS for Windows, version 22.0. Differences between experimental groups were evaluated using analysis of variance (ANOVA), with a significance level set at $p < 0.05$. Spearman correlation analysis was conducted to examine the relationship between the microbial community and environmental factors. The Modified Gompertz model was applied to fit the methane yield data.

3. Results and discussion

3.1 Characteristics of the digestate

The pH is an essential indicator of AD, with an optimal range for methanogenic activity being 6.5–8.0; deviations from this range can adversely affect methanogen performance [19]. Under mesophilic conditions, the pH initially decreased, then increased during the initial stage of AD, and ultimately stabilized between 7.5 and 7.7 (Figure 1a). Under thermophilic conditions, pH increased during the initial stage and eventually stabilized between 8.0 and 8.1 (Figure 1b). The enhanced

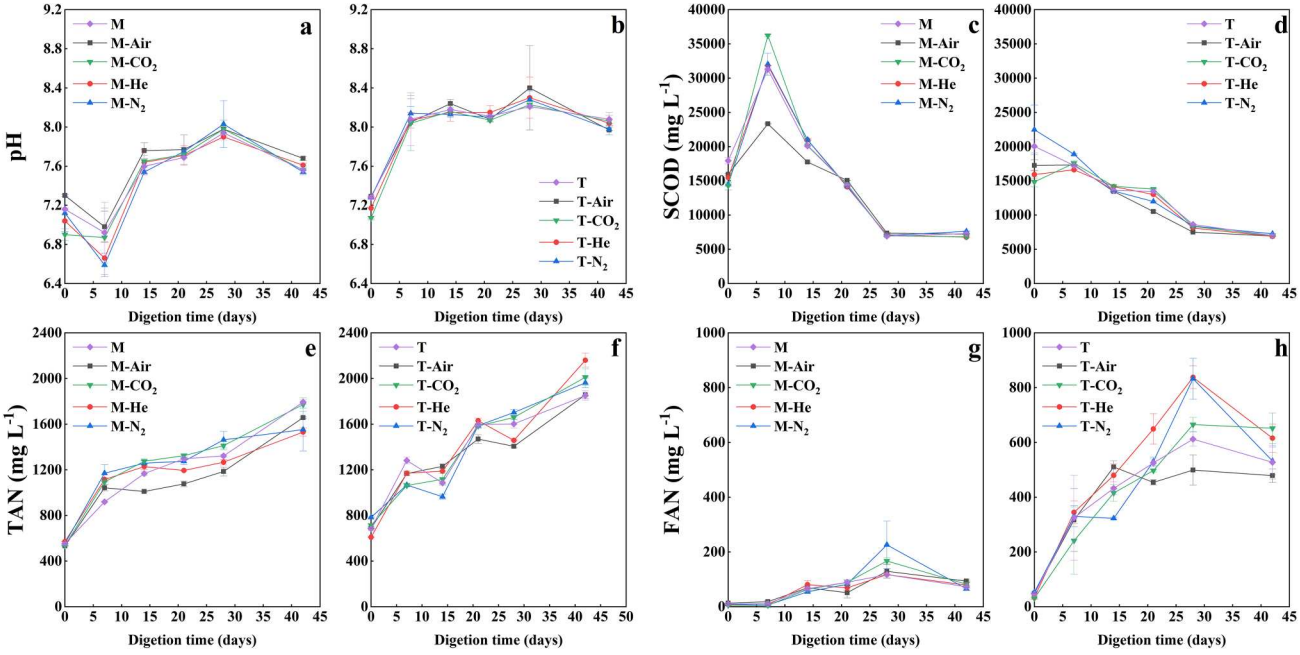


Figure 1. Changes of pH (a and b), SCOD (c and d), TAN (e and f) and FAN (g and h) during the co-digestion process under mesophilic and thermophilic conditions.

hydrolytic capacity under thermophilic conditions in AD releases more ammonia nitrogen, while the accelerated reaction rate leads to faster VFA consumption, resulting in a higher pH under thermophilic conditions compared with mesophilic conditions [3]. The addition of CO_2-NBW lowered the pH of both mesophilic and thermophilic digesters. The pH was similar to that of the CW and CM co-digestion systems with $NaHCO_3$ addition, but the digestate from AD with NBW addition was more advantageous for land use [3].

SCOD reflects the quantity of organic matter in digesters and the processes by which microorganisms produce or consume SCOD in AD [20]. Under mesophilic conditions, SCOD increased rapidly due to hydrolysis and subsequently decreased as methane production progressed (Figure 1c). Under thermophilic conditions, SCOD decreased continuously throughout the digestion process (Figure 1d). After day 28, all digesters reached a steady state in terms of SCOD due to the depletion of organic matter, with no significant differences observed among the digesters at this stage. At the end of AD, SCOD remained at a certain concentration, likely due to the presence of non-bioavailable organic matter, such as humic acids, within the digester [21]. Throughout AD, SCOD in all digesters were lower than those reported in previous study [3]. This reduction could be attributed to the addition of NBW or distilled water (DW), which likely enhanced mass transfer efficiency and improved organic matter utilization, as well as to the dilution effect [22,23].

Ammonia is typically produced through the hydrolysis of proteins in the substrate, TAN and FAN were higher under thermophilic conditions than under mesophilic ones [3] (Figure 1e–h). This is primarily due to the enhanced hydrolytic capacity under thermophilic conditions [9]. During AD, both TAN and FAN exhibited an increasing trend; however, methanogenic activity was inhibited when TAN levels exceeded 1500 mg L^{-1} or FAN levels exceeded 250 mg L^{-1} [3,9]. But ammonia inhibition in AD is typically influenced by multiple factors [24]. Based on the trend of SCOD, ammonia nitrogen in digesters did not negatively impact methane production, nor was there evidence of acid accumulation. TAN was greater but FAN was lower compared with previous study [3]. This suggests that NBW promoted the activity of hydrolysis enzymes to release more ammonia nitrogen, while lowering the pH to alleviate the inhibition of FAN to improve the stability of the system.

VFA are produced during the hydrolysis-acidification stage and consumed during the methanogenesis stage, serving as crucial intermediate products in the AD process [3,19]. Consequently, VFA initially increased and then decreased in the digesters (Figure 2). Under mesophilic conditions, VFA initially increased and then decreased. In the initial stage of mesophilic digestion, acetic acid predominated, while in the later stage, propionic acid became the dominant VFA. This shift is closely related to the Gibbs free energy of the various VFAs and the lower hydrogen partial pressure required

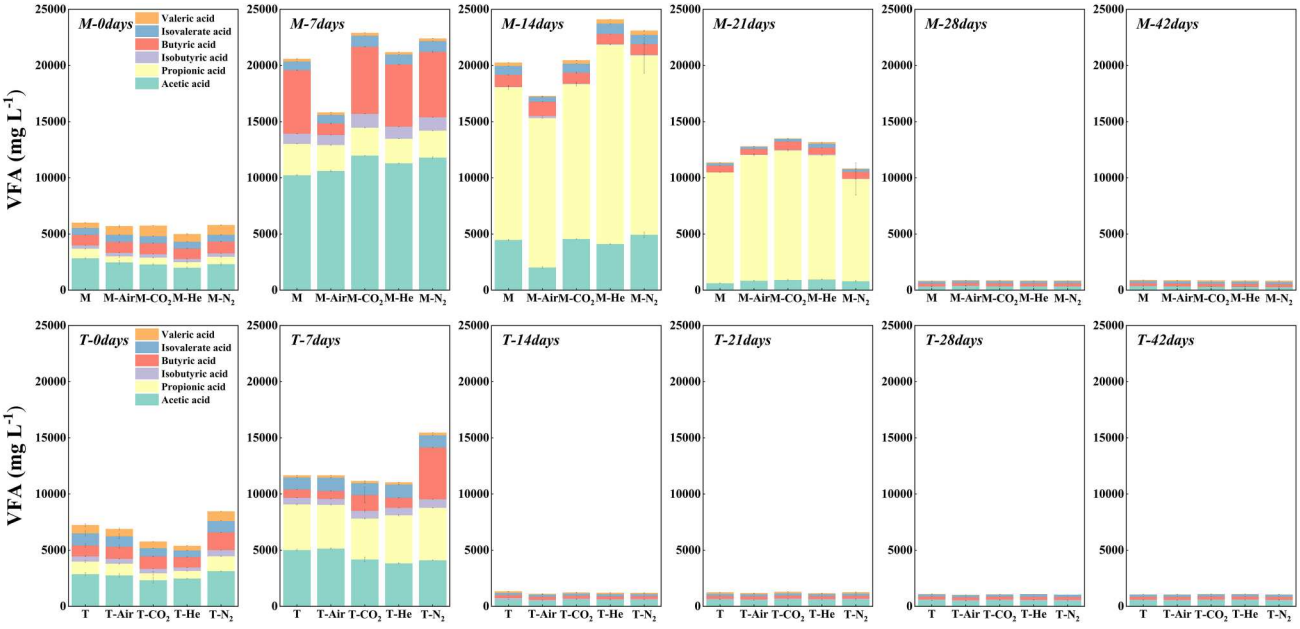


Figure 2. Changes of VFA during the co-digestion process under mesophilic and thermophilic conditions.

for the degradation of propionic acid [21]. The VFA of the digesters with added NBW was higher than that of the digesters with added DW, suggesting that NBW can promote hydrolysis and acidification [25]. On day 7, the butyric acid in the digesters with Air-NBW was lower than in the other digesters. Under thermophilic conditions, VFA increased and then decreased, with minimal variation after day 14. The rates of VFA production and consumption in the thermophilic digesters were higher than those in the mesophilic digesters, consistent with previous study [3]. The high temperature enhanced the hydrolysis of AD and the dissolution of organic matter, which increased the digestion rate of AD [9]. The VFA trends in mesophilic and thermophilic digesters were consistent with previous study, but the concentrations observed during AD were notably higher in both mesophilic and thermophilic digesters [3]. This increase may be attributed to differences in the AD substrate characteristics and the addition of NBW, which enhances hydrolysis [13].

TA (Figure 3a,b) serves as an indicator of the buffering capacity of the AD system, which is mainly provided by ammonia nitrogen [8]. The TA trends indicated no significant differences among the experimental groups across varying temperatures when different types of NBW were added. The TA was lower than the results of the previous study, but the previous study added $NaHCO_3$ [3]. The ratio of total volatile fatty acids to TA (TVFA/TA) (Figure 3c,d) is used to assess the stability of AD, with values exceeding 0.4 generally indicating instability [3,6]. In all digesters, TVFA/TA initially increased and then decreased, reflecting the production of VFAs during hydrolysis and acidification and their subsequent consumption during methanogenesis. In contrast, the thermophilic TVFA/TA was lower than that of the mesophilic digesters and stabilized more quickly. This is attributed to the faster reaction rates under thermophilic conditions, which led to more rapid VFA consumption [3]. The TVFA/TA for Air-NBW under mesophilic conditions was lower than that of the other groups during the mid-stage of digestion, indicating that Air-NBW enhanced VFA consumption and improved system stability [9,23]. The TVFA/TA ratio in the thermophilic digesters with NBW was not significantly different from that of the control digesters, possibly because the thermophilic predominantly influences hydrolysis, thereby diminishing the additional effects of NBW [26,27].

3.2 Methane yield

A modified Gompertz model was employed to fit the cumulative methane yield (CMY) (Figure 4 and Table 2). The fitting results showed a satisfactory agreement with the experimental results ($R^2 \geq 0.9936$). Under mesophilic or thermophilic conditions, the methane content of biogas did not differ significantly between experimental groups, ranging from 56.06% to 58.34%. Compared with thermophilic conditions, the addition of NBW significantly enhanced methane yield under mesophilic conditions.

Under mesophilic conditions, the ultimate methane yield (UMY) was obtained in the digesters added with

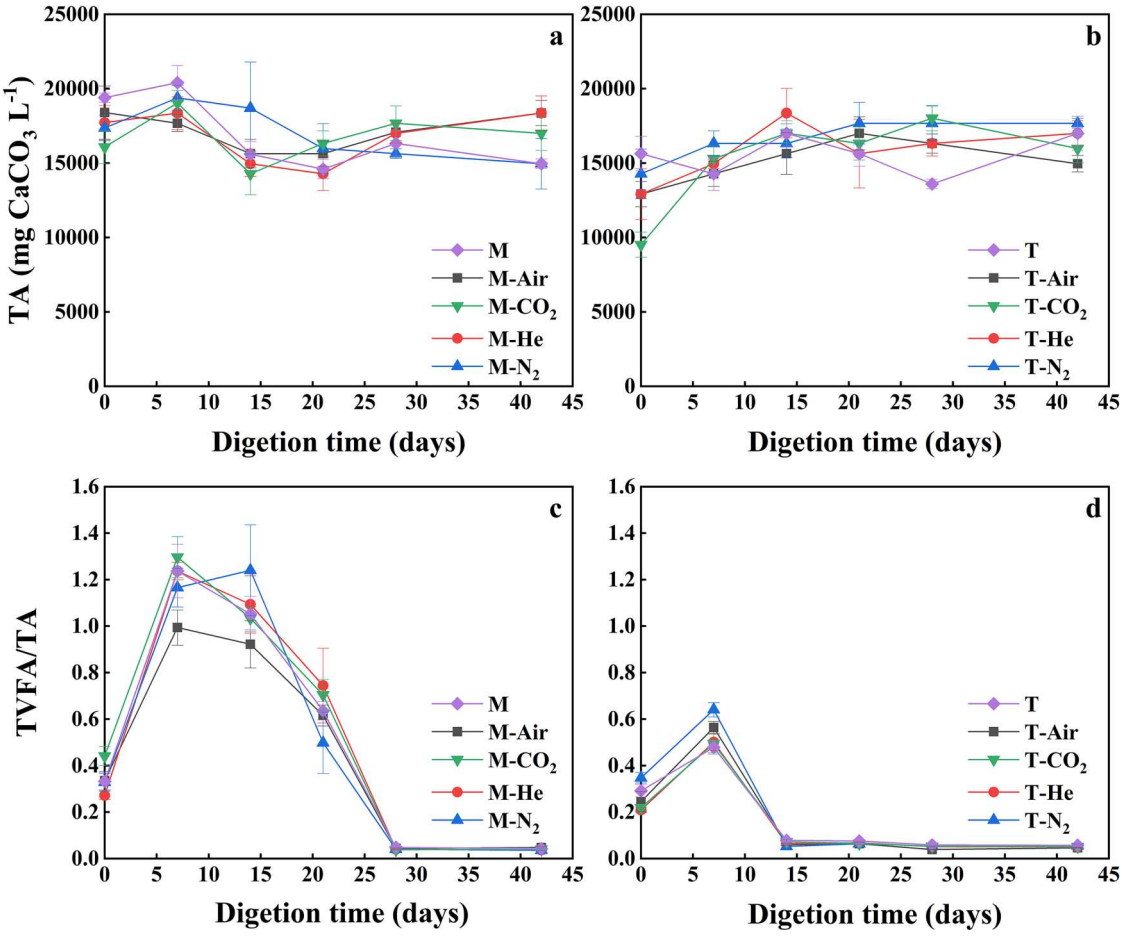


Figure 3. Changes of TA (a and b) and TVFA/TA (c and d) during the co-digestion process under mesophilic and thermophilic conditions.

NBW following descending order of 236.92 (M-Air) > 231.23 (M-CO_2) > 218.24 (M-He) > 201.56 (M) > 178.84 (M-N_2) mL g^{-1}-VS_{added}, M-Air, M-CO_2 and M-He were 17.54%, 14.72% and 8.28% higher than M, respectively. The methane yield of M-Air and M-CO_2 was significantly ($p < 0.05$) increased compared with M. The addition of iron filings to the co-digestion of barley and swine manure has been shown to enhance methane yield by 8.7% to 28.6% [28]. Similarly, the addition of cow manure-based biochar in mesophilic AD resulted in a 32.2% increase in methane yield [29]. However, the production of biochar and the addition of iron filings contribute to increased treatment costs and pose limitations for land application. The addition of Air-NBW facilitates the creation of a micro-oxygen environment during the hydrolysis stage of AD [30]. This environment enhances the activity of hydrolytic and acidogenic bacteria, thereby promoting the degradation of organic matter, which ultimately results in M-Air achieving the highest UMY [14]. The addition of CO_2-NBW also contributed to an increase in UMY under mesophilic conditions, likely due to the addition of CO_2, which enhanced the activity of hydrogenotrophic methanogens and bolstered system stability [25].

The addition of Air-NBW and CO_2-NBW significantly enhanced methane yield in the mesophilic digesters. However, previous study reported methane yield increases of 20.8% and 19.1%, respectively, which are higher than the gains observed in this experiment [23]. This discrepancy may be due to the high cellulose content in the CM, which likely impeded hydrolysis [23]. The addition of NBW mitigated the sharp drop in pH by acting as a buffer, accelerating the production and consumption of reducing sugars and VFA, and the addition of NBW simultaneously enhances the microbial activity of the rich microorganisms contained in the CM and promoted methane yield [22,31].

Under thermophilic conditions, the UMY was obtained in the digestors added with NBW following a descending order: 276.89 (T-N_2) > 265.03 (T-Air) > 262.18 (T) > 245.67 (T-CO_2) > 215.51 (T-He) mL g^{-1}-VS_{added}. T-N_2 and T-Air exhibited methane yield that were 5.61% and 1.09% higher than T, respectively. The proportional increase in methane yield with NBW

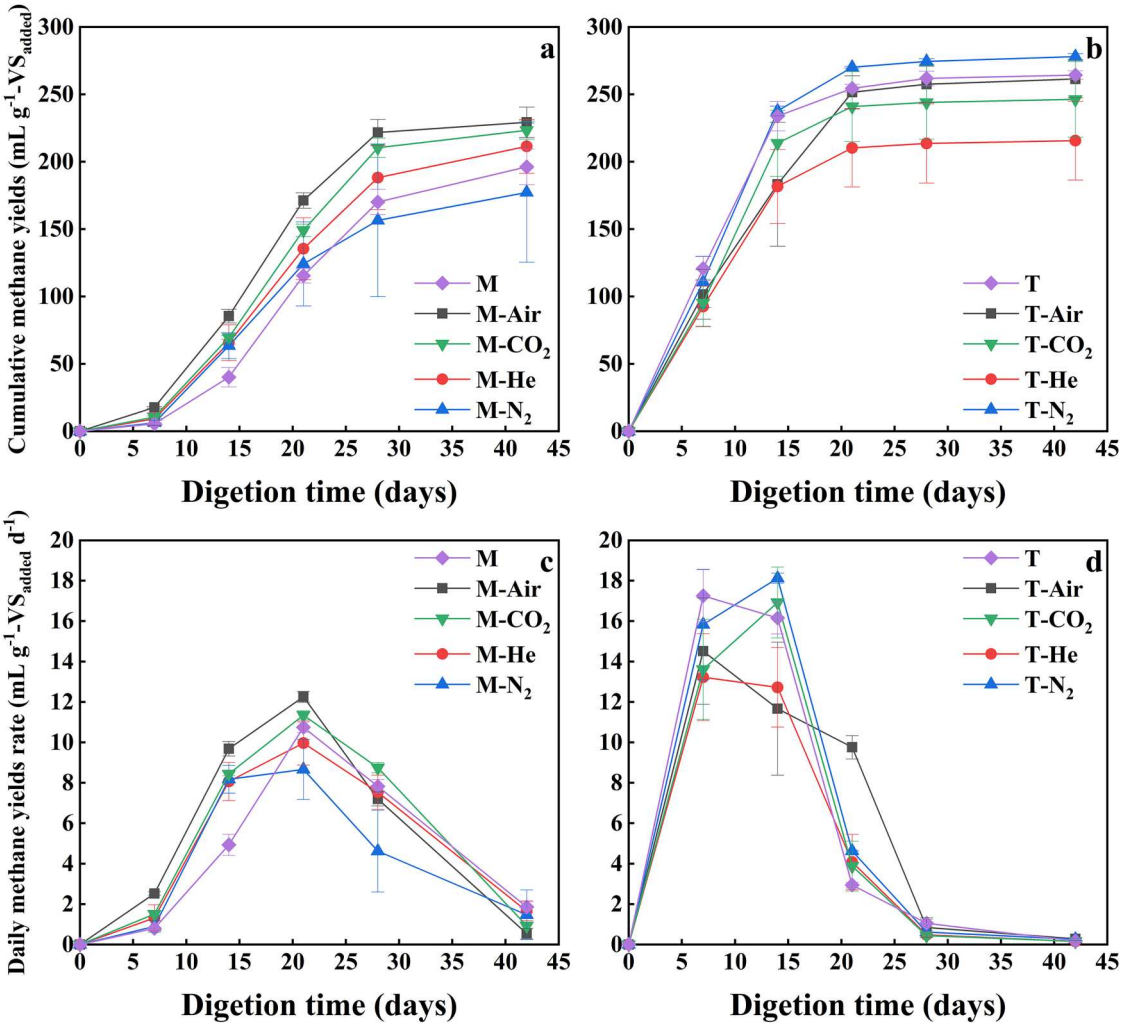


Figure 4. Changes of cumulative methane yield (a and b) and methane yield rate (c and d) during the co-digestion process under mesophilic and thermophilic conditions.

addition under thermophilic conditions was less pronounced compared with the 35.7% increase observed with cow manure biochar, likely due to the reduced NBW concentration resulting from its volatilization under thermophilic conditions [29]. This phenomenon occurs because NBW concentrations in water diminish as temperature rises[26,27]. The methane yield in the T-N_2 was significantly higher than in the T ($p < 0.05$). The highest UMY, 276.89 mL g^{-1}-VS_{added}, was achieved in the T-N_2. This enhancement is attributed to the increased hydrolysis of proteins at high temperature, while adding N_2 improves the degradation of high molecular weight compounds (polysaccharides and proteins) [9,16].

Table 2. Parameters of modified Gompertz model.

	Ultimate methane yield, UMY (mL g^{-1} VS_{added})	Maximum methane yield rate, R_m (mL g^{-1} VS_{added} d^{-1})	Lag phase, λ (d)	R^2
M	201.56 ± 13.31	11.51 ± 2.54	10.63 ± 3.34	0.9993
M-Air	236.92 ± 11.34	14.07 ± 0.85	7.68 ± 0.05	0.9972
M-CO_2	231.23 ± 6.69	13.22 ± 1.53	8.74 ± 1.53	0.9972
M-He	218.24 ± 19.93	11.28 ± 3.32	8.25 ± 0.28	0.9993
M-N_2	178.84 ± 31.65	10.03 ± 2.55	7.80 ± 0.36	0.9997
T	262.18 ± 21.62	26.10 ± 3.51	2.37 ± 2.55	0.9997
T-Air	265.03 ± 14.78	16.80 ± 1.53	1.38 ± 1.14	0.9936
T-CO_2	245.67 ± 12.44	24.80 ± 0.34	3.16 ± 0.51	1.0000
T-He	215.51 ± 5.83	18.75 ± 0.72	2.10 ± 0.25	0.9997
T-N_2	276.89 ± 11.73	26.14 ± 2.16	2.76 ± 2.38	0.9999

Average and standard deviation were calculated based on three parallels.

The rate of methane yield in the mesophilic digesters (Figure 4c) initially increased and then declined, peaking on day 21. The mesophilic digesters added with NBW exhibited higher initial rate of methane yield. This finding aligns with previous research indicating that NBW supplementation enhances the rate of methane yield [23,30]. Under thermophilic conditions, the rate of methane yield (Figure 4d) also rose initially and then declined. In contrast to the mesophilic digesters, the thermophilic digesters achieved its maximum methane yield on day 7 and day 14, which occurred earlier than in the mesophilic digesters. Compared with mesophilic conditions, thermophilic conditions enhanced the solubility of organic matter, accelerated both biological and chemical processes, and reduced the retention time [3]. The fitting results indicated that the mesophilic digesters with Air-NBW and CO_2-NBW additions had a shorter lag phase, and the thermophilic digesters with Air-NBW had an even shorter lag phase. The lag phase in the thermophilic digesters were significantly shorter compared with the mesophilic digesters. This confirms that higher temperatures accelerate biological and chemical reaction rates [3].

3.3 Dominant microbial community

3.3.1 Alpha diversity in bacteria and archaea

Observed species (sobs) represent the number of detected OTUs, with larger values indicating higher diversity. A higher Chao1 value reflects greater microbial diversity in the digesters [3]. Both the sobs and Chao1 for bacteria in the mesophilic and thermophilic digesters with NBW addition were higher than those in the control group (Table S1), suggesting that NBW enhances bacterial diversity in the digesters. Simpson and Shannon, on the other hand, reflect the microbial diversity in the samples [25]. Therefore, the bacterial abundance in the mesophilic and thermophilic digesters with NBW addition was higher compared with the control. In terms of alpha diversity for archaea (Table S2), the Chao1 in the mesophilic digesters with He-NBW and N_2-NBW additions were lower than those in the control, indicating reduced archaeal diversity compared with the control group. The archaeal diversity in the thermophilic digesters with Air-NBW were lower than in the control. However, based on the Shannon and Simpson, both mesophilic and thermophilic digesters with NBW exhibited greater archaeal diversity compared with the control. A previous study reported that digesters with Air-NBW and CO_2-NBW additions had higher OTUs and Chao1 for both bacteria and archaea [32].

3.3.2 Dominant bacterial community

Under mesophilic conditions, the bacterial community was predominantly dominated by *Firmicutes* and *Bacteroidota* on both day 6 and day 28 (Figure 5). *Firmicutes* and *Bacteroidota* are capable of degrading organic matter into VFA, H_2 and CO_2, which are utilized by methanogens [33]. Additionally, *Firmicutes* serve as indicators of AD stability. *Firmicutes* exhibited a significant negative correlation with TA ($p < 0.01$, $R = -0.7118$), propionic acid ($p < 0.001$, $R = -0.8838$), and butyric acid ($p < 0.01$, $R = -0.6874$) (Figure S2). The addition of Air-NBW and He-NBW increased the relative abundance (RA) of *Bacteroidota* under mesophilic conditions. *Bacteroidota* exhibited a higher RA in digesters with the addition of NBW [32]. *Bacteroidota* demonstrated a significant positive correlation with TA ($p < 0.001$, $R = 0.7665$) and propionic acid ($p < 0.001$, $R = 0.7856$). *Proteobacteria* are capable of metabolizing complex organic matter to produce acetic acid [34]. The higher RA of both *Proteobacteria* and *Bacteroidota* suggests that the digestion of acetic acid, leading to methane yield, is more likely to occur in the digesters [34]. The RA of *Proteobacteria* appeared to decrease as digestion progressed. The addition of Air-NBW, He-NBW, and CO_2-NBW to the mesophilic digesters effectively increased the RA of *Proteobacteria*. *Proteobacteria* exhibited a significant positive correlation with UMY ($p < 0.01$, $R = 0.7092$).

Under thermophilic conditions, the bacterial community was predominantly dominated by *Firmicutes* on both day 6 and day 28 (Figure 5). The thermophilic digesters with different NBW amendments exhibited greater bacterial abundance on day 6 compared with those with DW. The addition of NBW enriched *Thermotogota* during the initial stage of AD (day 6). A decline in the RA of Thermotogota was observed in the later phase of AD (day 28) within the He-NBW and N_2-NBW amended digesters. *Thermotogota* is a thermophilic bacterium responsible for fibre degradation and VFA production in thermophilic digesters [35]. *Thermotogota* exhibited a significant negative correlation with TAN ($p < 0.01$, $R = -0.8103$) and a significant positive correlation with acetic acid ($p < 0.01$, $R = 0.7687$) (Figure S3).

3.3.3 Dominant archaeal community

At the genus level, *Methanosarcina*, *Methanobacterium*, *Methanosphaera*, *Methanothermobacter*, and *Methanobrevibacter* dominated in the mesophilic digesters during the initial stage, with *Methanobrevibacter* exhibiting the highest RA (Figure 6). In the late stage of digestion, *Methanosarcina* and *Methanobacterium* emerged as the top two genera in terms of RA. The addition of Air-

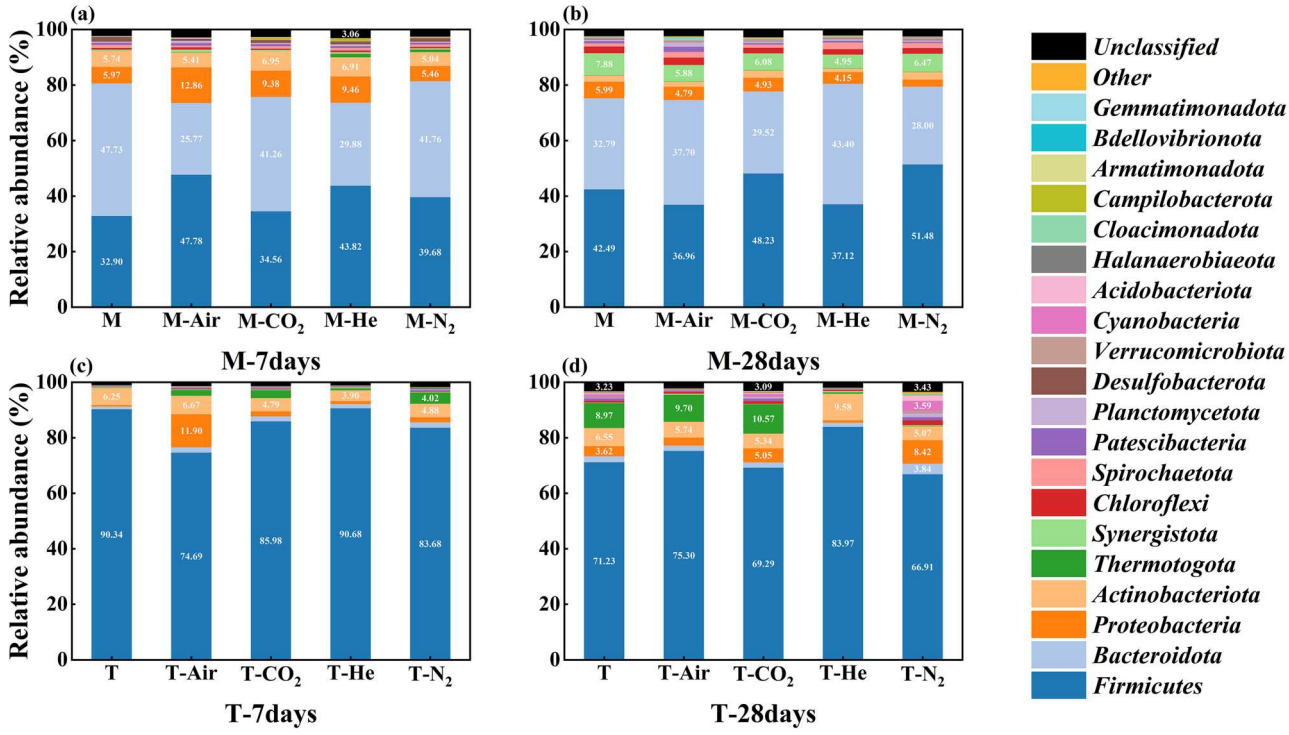


Figure 5. Relative abundance of bacterial community at the phylum level.

NBW promoted the enrichment of *Methanosarcina* during the initial stage of digestion, as these mixotrophic methanogens can utilize acetic acid and H_2/CO_2 to produce methane [36]. *Methanosarcina* exhibited a positive correlation with UMY ($p < 0.05$, $R = 0.5565$) (Figure S2). The addition of He-NBW, N_2-NBW, and CO_2-NBW promoted the enrichment of the hydrogenotrophic methanogen *Methanobacterium* during the late stage of digestion [37]. *Methanobacterium* exhibited a higher RA in the digesters with NBW addition, unlike a

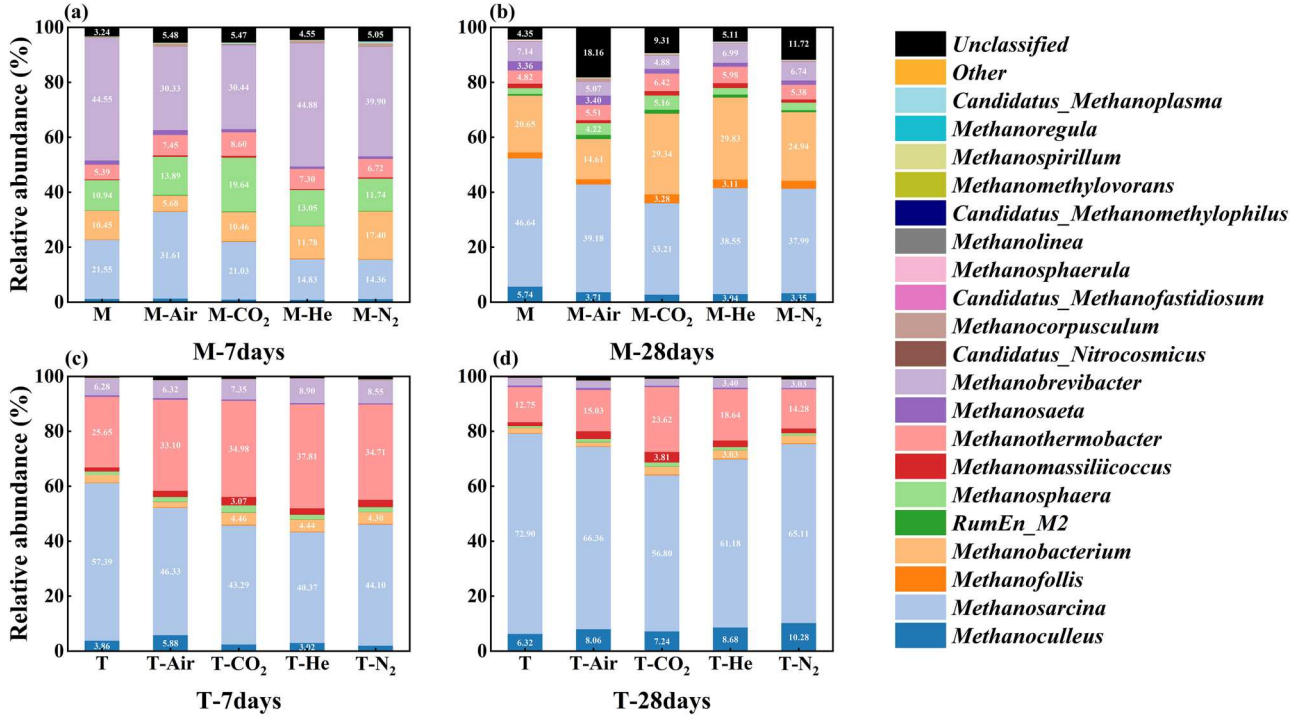


Figure 6. Relative abundance of archaeal community at the genus level.

previous study, the RA of *Methanobacterium* was lowest in the mesophilic digesters with Air-NBW in this experiment [32,33]. *Methanobacterium* was significantly positively correlated with TVFA ($p < 0.01$, $R = 0.6874$), indicating that it is more tolerant to unfavourable environments as a hydrogenotrophic methanogen. *Methanosphaera* utilizes acetic acid as a carbon source and hydrogen to reduce methanol, resulting in methane yield. An acetic acid-rich environment fosters the growth of *Methanosphaera*, but as acetic acid concentrations decline in the later stages of digestion, the RA of *Methanosphaera* correspondingly decreases [38]. *Methanosphaera* exhibited a significant positive correlation with TAN ($p < 0.01$, $R = 0.6552$) and UMY ($p < 0.01$, $R = 0.6656$).

Methanosarcina and *Methanothermobacter* were the predominant archaeal genera in the thermophilic digester on both day 6 and day 28 (Figure 6). The addition of NBW enhanced the RA of *Methanothermobacter* and *Methanobrevibacter*, although their RA declined as digestion progressed. As digestion progressed, the RA of *Methanosarcina* and *Methanoculleus* increased, with NBW addition particularly favouring the rise in RA of *Methanoculleus* during the later stage. *Methanosarcina*, being a mixotrophic methanogen, can utilize acetic acid and CO_2/H_2 for methane yield and is notably tolerant to high ammonia concentration [36]. *Methanothermobacter, Methanobrevibacter, and Methanoculleus* are hydrogenotrophic methanogens [37,39]. Consequently, the addition of NBW effectively promoted the enrichment of hydrogenotrophic methanogens. Among them, *Methanoculleus* was considered the most effective microorganism at the archaea, which significantly impacts methane yield and can tolerate high ammonia concentration [39]. *Methanoculleus* exhibited a significant positive correlation with TAN ($p < 0.01$, $R = 0.7455$) (Figure S3).

4. Conclusion

The addition of NBW enhanced the hydrolysis of CW and CM during co-digestion under both thermophilic and mesophilic conditions, while also sustaining a stable methane yield even at elevated TAN level. Under mesophilic conditions, methane yields increased by 17.54%, 8.28%, and 14.72% with the addition of Air-NBW, He-NBW, and CO_2-NBW, respectively. Similarly, under thermophilic conditions, methane yields increased by 1.09% and 5.61% with the addition of Air-NBW and N_2-NBW, respectively. The addition of NBW also enhanced microbial abundance and diversity in both mesophilic and thermophilic digesters. NBW presents a promising alternative to distilled water for AD processes requiring dilution; however, further research is needed to elucidate the mechanisms through which NBW affects AD and to assess the impact of other potential contaminants.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data presented in this study are available on request from the authors.

Author contributions

Youfei Zhou: Methodology, Validation, Formal analysis, Investigation, Formal analysis, Data curation, Writing-original draft. **Peilin Guo:** Methodology, Validation, Formal analysis, Investigation, Formal analysis, Data curation, Writing-review & editing. **Yi Liu:** Methodology, Validation, Formal analysis, Investigation, Formal analysis, Data curation. **Weijie Hu:** Methodology, Validation, Formal analysis, Investigation, Formal analysis, Data curation. **Tianfeng Wang:** Methodology, Validation, Investigation, Formal analysis, Resource, Funding acquisition.

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